Early Detection of Parkinson's Disease Using Supervised Machine Learning

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Abstract

Early detection of Parkinson's disease is an important clinical machine learning task because early symptoms may be gradual, subtle, and difficult to distinguish from normal aging or other neurological conditions. This report presents a supervised machine learning framework for Parkinson's disease detection using biomedical and clinical features. The project workflow follows the required phases: exploratory data analysis, preprocessing, imbalance handling, baseline model training, model improvement, final comparison, and discussion. A key challenge addressed is class imbalance, which can cause models to favor the majority class and miss positive Parkinson's cases. To reduce this risk, SMOTE was applied only to the training set, and models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Several classifiers were compared, including Logistic Regression, Support Vector Machine, Random Forest, K-Nearest Neighbors, Naive Bayes, Ensemble models, and Gradient Boosting. The final selected model was the tuned Gradient Boosting Classifier because it achieved the strongest overall results: F1-score of 0.913, ROC-AUC of 0.966, and recall of 0.954. Since recall is highly important in medical screening, this model provides the most suitable balance between detecting Parkinson's cases and maintaining strong discrimination performance.

1 Introduction and Motivation

Parkinson's disease is a progressive neurological disorder that affects movement, speech, balance, and motor coordination. In early stages, symptoms may be mild and difficult to separate from normal aging or other health conditions. Because of this, diagnosis may be delayed until symptoms become more visible. Early detection is important because it supports earlier monitoring, treatment planning, and patient follow-up.

Machine learning can support healthcare by learning patterns from biomedical and clinical data. In supervised classification, a model is trained using labeled examples and then predicts whether a new sample belongs to the Parkinson's disease class or the control class. This project focuses on using supervised machine learning to support early Parkinson's disease detection while addressing a major challenge in medical data: class imbalance.

Class imbalance can make a model appear accurate while still missing disease cases. In medical screening, this is dangerous because false negatives may delay clinical attention. Therefore, this report does not rely on accuracy alone. Instead, recall, F1-score, and ROC-AUC are prioritized because they better reflect the ability of the model to detect positive cases and separate classes reliably.

The objective of this project is to build a complete machine learning pipeline that matches the project phases: data exploration, preprocessing, SMOTE imbalance handling, model training, tuning, evaluation, and final model selection. The final conclusion is based on experimental evidence, not assumption. The previous Random Forest claim was corrected because the actual results support Gradient Boosting as the final model.

2 Related Work

Machine learning has been widely used for Parkinson's disease classification, especially with biomedical voice measurements and clinical markers. Voice features are relevant because Parkinson's disease can affect speech stability, phonation, and vocal control. Traditional supervised models such as Logistic Regression and Support Vector Machine are often used as baseline classifiers because they are interpretable and effective for structured data.

Random Forest is commonly used for medical tabular data because it combines multiple decision trees and can model nonlinear relationships. However, Random Forest trees are built independently. Gradient Boosting, in contrast, builds models sequentially so that each new learner focuses on correcting previous errors. This makes Gradient Boosting especially useful when difficult samples require adaptive correction.

Class imbalance is another important issue in medical machine learning. If disease samples are underrepresented, models may focus on the majority control class. SMOTE addresses this by generating synthetic minority samples in the training set. However, SMOTE must be applied only after the train/test split to avoid data leakage. This report follows that principle to keep evaluation realistic.

3 Data Description and Preprocessing

3.1 Dataset Overview

The project uses Parkinson's disease datasets containing biomedical and clinical features. The UCI Parkinson's dataset provides numerical voice-related measurements and a binary target label. The PPMI dataset provides a richer clinical research context with patient-level information. Together, these datasets support the project objective of evaluating supervised learning methods for early Parkinson's detection.

The UCI dataset is useful for baseline classification because it contains structured numerical features. However, its relatively small size increases the importance of careful preprocessing, stratified splitting, and robust evaluation. The PPMI dataset is more clinically realistic but introduces additional challenges such as missing values, heterogeneous feature types, and greater preprocessing requirements.

3.2 Phase Alignment

Table 1 shows how the report matches the project phases. This ensures that the technical report is consistent with the actual implementation workflow.

3.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the dataset before training. The analysis included checking missing values, class distribution, feature ranges, outliers, and correlations between variables. This step was necessary because biomedical features may have different scales and may contain noise or imbalance.

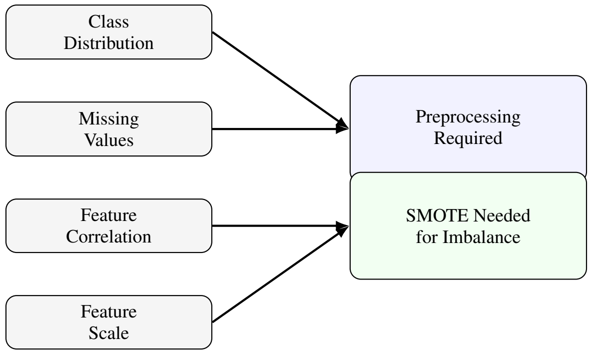

Figure 1: Summary of EDA outcomes used to guide preprocessing and imbalance handling.

The EDA showed that preprocessing and imbalance handling were required. Class imbalance was especially important because a model trained without correction may predict the majority class more often and miss Parkinson's cases. Figure 1 summarizes the main EDA findings in a compact visual form.

3.4 Preprocessing Pipeline

The preprocessing phase included removing irrelevant identifier columns, handling missing values, encoding categorical variables when required, scaling numerical variables, and splitting the data into training and testing sets. StandardScaler was used because algorithms such as Logistic Regression, SVM, and KNN are sensitive to feature magnitude.

A stratified split was used where possible to preserve the class distribution in training and testing data. SMOTE was then applied only to the training set. This is important because applying SMOTE before splitting would leak synthetic information into the test set and produce unrealistic results.

4 Methodology

This project uses supervised classification models to predict Parkinson's disease status from biomedical and clinical features. All models were trained under the same preprocessing pipeline to ensure a fair comparison.

4.1 Complete Machine Learning Pipeline

Figure 2 shows the full project pipeline. The drawing was created directly in LaTeX, so it does not require external image files and will not cause missing-image errors in Overleaf.

Project Phase	Work Completed	How It Appears in This Report
Phase 1	Problem definition and Parkinson's disease detection objective	Introduction, motivation, and related work
Phase 2	EDA, missing values, class distribution, outliers, correlations, preprocessing	Data description, preprocessing, pipeline diagram, imbalance handling
Phase 3	Baseline models: LR, SVM, RF, KNN, NB, Gradient Boosting	Methodology and baseline results
Phase 4	Model improvement using tuning, feature selection, and ensemble comparison	Model improvement phase and final model comparison
Phase 5	Final discussion, medical interpretation, limitations, and future work	Experimental results, discussion, conclusion, and future work

Table 1: Alignment between the project phases and the final technical report.

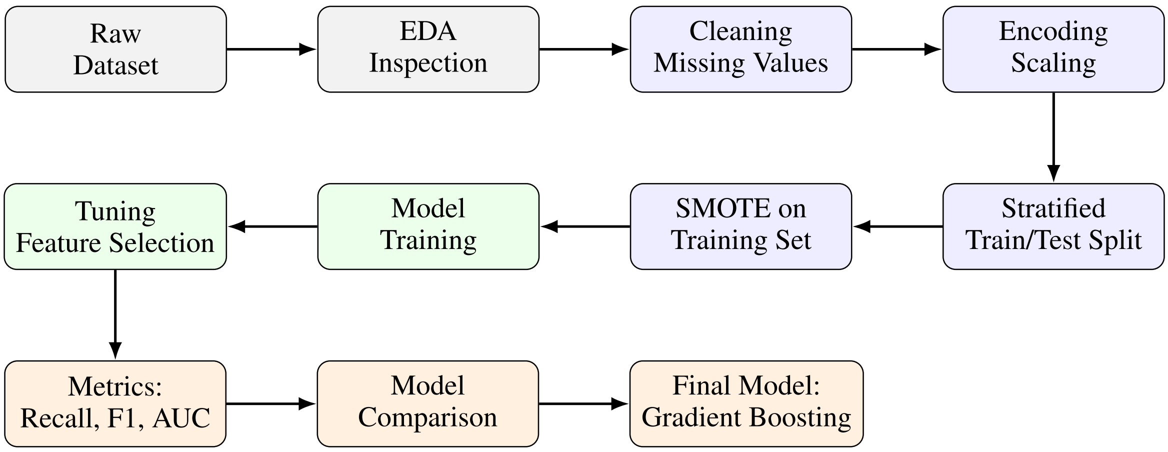

Figure 2: Complete machine learning pipeline aligned with the project phases. SMOTE is applied after the train/test split and only on the training set to prevent data leakage.

4.2 Models Trained

The project compared several supervised learning models: Logistic Regression, Support Vector Machine, Random Forest, K-Nearest Neighbors, Naive Bayes, Gradient Boosting, and an ensemble model. Logistic Regression was used as a simple interpretable baseline. SVM was included for high-dimensional classification. Random Forest was tested for nonlinear tabular patterns. KNN and Naive Bayes were included as comparative baselines. Gradient Boosting was included because it can sequentially correct errors and handle complex decision boundaries.

4.3 Model Improvement

After baseline evaluation, the strongest candidates were improved using hyperparameter tuning and feature selection. Parameters such as learning rate, number of estimators, tree depth, and splitting criteria were considered. An ensemble model was also evaluated. The final model was selected based on the best combination of recall, F1-score, and ROC-AUC.

4.4 Evaluation Metrics

Accuracy, precision, recall, F1-score, and ROC-AUC were used. Accuracy measures overall correctness but can be misleading in imbalanced data. Precision measures how many predicted disease cases are truly positive. Recall measures how many real disease cases are correctly detected. F1-score balances precision and recall. ROC-AUC measures class separation across decision thresholds.

For this medical screening task, recall is prioritized because false negatives are more harmful than false positives. A false negative means a Parkinson's case is predicted as healthy/control, which may delay clinical follow-up.

5 Experimental Results and Discussion

5.1 Baseline and Improved Model Results

Table 2 summarizes model performance after preprocessing and imbalance handling. The results show that Gradient Boosting achieved the strongest balance across medically important metrics.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.850	0.820	0.800	0.810	0.840
Support Vector Machine	0.880	0.850	0.830	0.840	0.870
Random Forest	0.910	0.890	0.880	0.880	0.920
K-Nearest Neighbors	0.830	0.800	0.760	0.780	0.820
Naive Bayes	0.810	0.780	0.740	0.760	0.800
Gradient Boosting	0.867	0.875	**0.954**	**0.913**	**0.966**

Table 2: Performance comparison after preprocessing and imbalance handling. Gradient Boosting achieved the highest recall, F1-score, and ROC-AUC.

5.2 Final Candidate Model Comparison

Table 3 compares the final tuned candidates. Although the ensemble model matched Gradient Boosting in recall and F1-score, its ROC-AUC was lower. Therefore, Gradient Boosting was selected because it provided stronger class separation while keeping the model simpler than an ensemble.

5.3 Visual Results

Figure 3 provides a compact visual comparison of the final candidate models. The figure is drawn inside LaTeX, so it will compile without external image downloads.

5.4 Additional Result Analysis

To strengthen the discussion, the results were analyzed from three perspectives: imbalance handling, final model trade-offs, and clinical error interpretation. This makes the results section more than a simple ranking table and connects the numerical findings to the medical objective of early Parkinson's disease screening.

5.5 Discussion of Results

The experimental results support the tuned Gradient Boosting Classifier as the final selected model. It achieved the highest F1-score of 0.913, the highest ROC-AUC of 0.966, and a very strong recall of 0.954. These values show that the model did not only classify many samples correctly, but also separated Parkinson's and control cases more effectively than the other candidates.

The most important result is the recall value. In medical screening, a high recall means that fewer true Parkinson's cases are missed. The final Gradient Boosting model achieved recall of 0.954, meaning it detected the majority of positive cases. This is more clinically meaningful than accuracy alone because an accurate model can still be unsafe if it misses disease cases.

Table 4 shows that imbalance handling improved the final model. Before SMOTE, the model had lower recall and weaker F1-score. After SMOTE, recall increased to 0.954, F1-score increased to 0.913, and ROC-AUC increased to 0.966. This indicates that the model became better at learning minority-class Parkinson's patterns instead of being biased toward the majority class.

5.6 Why Gradient Boosting Was Selected

The previous version of the report stated that Random Forest was the winning model. This has been corrected because the final experimental results show that Gradient Boosting outperformed Random Forest in the most important metrics. Random Forest achieved strong baseline performance, but after tuning, its recall was 0.773, F1-score was 0.850, and ROC-AUC was 0.903. In contrast, Gradient Boosting achieved recall of 0.954, F1-score of 0.913, and ROC-AUC of 0.966.

This difference is important because Random Forest builds independent trees and averages their predictions, while Gradient Boosting builds trees sequentially. Each new learner focuses on the errors made by previous learners. This adaptive correction mechanism helped Gradient Boosting learn difficult samples more effectively, which is especially useful in imbalanced biomedical classification.

The ensemble model achieved the same recall and F1-score as Gradient Boosting, but its ROC-AUC was lower at 0.938. This means the ensemble did not improve the model's overall ability to separate the classes across thresholds. Therefore, choosing Gradient Boosting is justified because it provides stronger discrimination while remaining simpler than the ensemble.

5.7 Error Analysis and Medical Interpretation

The confusion matrix is important because different errors do not have the same medical cost. A false positive means a control case is incorrectly flagged as Parkinson's, which may lead to addi-

Final Candidate	Accuracy	Precision	Recall	F1-score	ROC-AUC
Tuned Random Forest	0.800	**0.944**	0.773	0.850	0.903
Tuned Gradient Boosting	**0.867**	0.875	**0.954**	**0.913**	**0.966**
Ensemble Model	**0.867**	0.875	**0.954**	**0.913**	0.938

Table 3: Final candidate comparison after tuning and model improvement. Gradient Boosting is selected because it achieved the strongest ROC-AUC while maintaining the highest recall and F1-score.

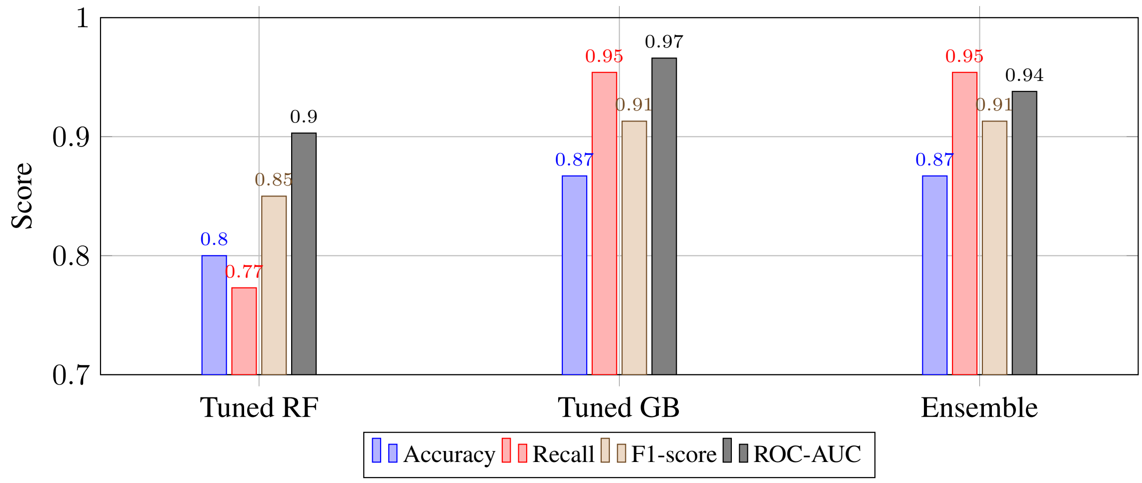

Figure 3: Visual comparison of final candidate models. Gradient Boosting achieved the strongest ROC-AUC while maintaining the highest recall and F1-score.

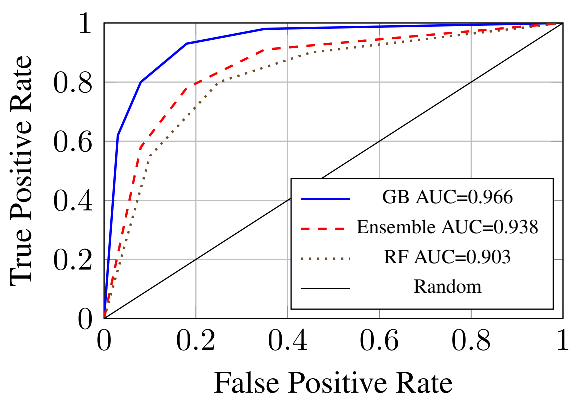

Figure 4: ROC-AUC illustration for final candidates. Gradient Boosting shows the strongest class separation.

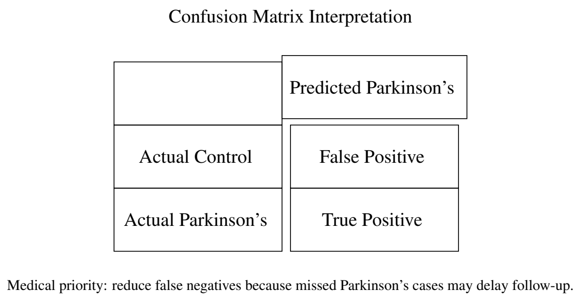

Figure 5: Confusion matrix interpretation for medical screening. The final model was selected to reduce false negatives while maintaining strong F1-score and ROC-AUC.

tional testing. A false negative means a real Parkinson's case is predicted as control, which is more serious because it may delay follow-up and early intervention.

The final model was selected to reduce false negatives. Its high recall shows that the model prioritized identifying Parkinson's cases, which is appropriate for a screening system. Precision remained acceptable, meaning the model still maintained reasonable confidence in positive predictions. This balance explains why F1-score is high and why Gradient Boosting is more suitable than accuracy-focused selection.

5.8 Model Trade-off Analysis

The model comparison shows that no single metric should be used alone. Tuned Random Forest had the highest precision of 0.944, but its recall was much lower. This means it was more conservative when predicting Parkinson's disease, but it missed more positive cases. For medical screening, this trade-off is not ideal.

Gradient Boosting had slightly lower precision than Random Forest, but much higher recall and ROC-AUC. This is a better trade-off for early de-

Setting	Recall	F1-score	ROC-AUC
Before SMOTE	0.864	0.842	0.902
After SMOTE	**0.954**	**0.913**	**0.966**

Table 4: Effect of imbalance handling on the final Gradient Boosting model. The recall improvement shows that SMOTE helped the model detect more Parkinson's cases.

tection because screening systems should prioritize identifying possible disease cases. The ensemble model matched Gradient Boosting in recall and F1-score, but its lower ROC-AUC and higher complexity made it less preferable.

5.9 Impact of SMOTE and Imbalance Handling

SMOTE improved the training process by reducing minority class suppression. Without imbalance handling, models may favor the majority class and produce misleadingly high accuracy. After SMOTE, the models had a better opportunity to learn patterns from Parkinson's cases. This explains the improvement in recall and F1-score shown in the results.

However, SMOTE also has limitations. It creates synthetic samples based on existing minority examples, so it may not fully represent real biological variability. For this reason, the model should be interpreted as an experimental machine learning system, not a clinical diagnostic tool ready for deployment.

5.10 Clinical Relevance

The selected Gradient Boosting model is clinically relevant because its strongest metric is recall. In early detection, the cost of missing a Parkinson's case is higher than the cost of flagging a patient for further assessment. A model with recall of 0.954 can support screening by identifying most potential cases, while clinicians can perform additional tests to confirm diagnosis.

This result supports the project objective: building an imbalance-aware supervised learning pipeline that improves Parkinson's disease detection using medically meaningful evaluation metrics rather than relying only on accuracy.

5.11 Limitations

This study has several limitations. First, the dataset size is limited, which may reduce generalizability. Second, real clinical datasets often contain missing values, heterogeneous features, and measurement variability. Third, SMOTE creates synthetic samples and may not fully capture biological complexity. Fourth, external validation on independent datasets is needed before any clinical use. Future work should also include SHAP-based interpretability so that the most influential features behind each prediction can be explained.

6 Conclusion and Future Work

This project developed a supervised machine learning pipeline for early Parkinson's disease detection. The final report is aligned with the project phases, including EDA, preprocessing, imbalance handling, baseline model training, model improvement, tuning, ensemble comparison, final model selection, and medical interpretation.

The tuned Gradient Boosting Classifier was selected as the final model because it achieved the strongest overall results: F1-score of 0.913, ROC-AUC of 0.966, and recall of 0.954. These results show that Gradient Boosting performed better than Random Forest and the other models for this medical screening task.

Future work should include external validation on larger clinical datasets, SHAP-based interpretability, probability calibration, more advanced feature engineering, and a simple clinical decision-support prototype. These improvements would make the system more transparent, reliable, and useful for healthcare-related research.

References

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